

Microeconomics Qualifier Playbook

Exam Patterns, Attack Protocols, and Proof Templates

Built from the sample microeconomics qualifier exams in the provided archive

Version 1.0

Purpose. This is not a textbook and not a solution manual. It is a methodological reference for taking practice qualifiers: how to recognize recurring problem types, what to write first, what theorem or proof move usually unlocks the problem, what picture to draw, and what mistakes to avoid.

Use rule. During practice, do not read the whole document passively. After attempting a problem, locate the matching protocol, compare your solution to the checklist, and update your personal mistake bank.

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Source Base and Scope

The document is based on recurring patterns in the provided sample qualifying exams, primarily ASU microeconomics qualifiers from 2013 through 2024. The exams repeatedly combine six topic families: partial equilibrium and consumer theory; decision theory and uncertainty; static and dynamic games; general equilibrium and welfare; Bayesian games and asymmetric information; and mechanism design. The playbook is organized by *exam action*, not by textbook chapter.

Exam period	Recurring content used to build this playbook
2013–2015	Owner-manager dynamic games, repeated/overlapping-generation games, insurance and credit screening, moral hazard, cost functions, Edgeworth boxes, direct revelation mechanisms, sequential games, Cournot existence, local nonsatiation, and expenditure/wealth representations.
2016–2018	Pure exchange and Robinson Crusoe economies, nontraded goods, Pareto optimality, input-demand monotonicity, signaling and WPBE, sequential equilibrium, quasilinear demand, no-Giffen arguments, Bayesian games, single crossing, value functions, closed-graph equilibrium correspondences, and uncertainty Edgeworth boxes.
2019–2021	Repeated games, advertising/core problems, moral hazard with MLRP/FOA, competitive screening, aggregate excess demand, adverse selection, abstract economy/auctioneer formulations, flexibility value, warranty screening, incomplete-information Cournot, verifiable disclosure, OLG prisoners' dilemma, and mean-variance portfolio choice.
2022–2024	Robinson Crusoe and welfare theorem edge cases, subjective expected utility testable implications, moral hazard lending, Bayesian disclosure, excess-demand WARP and convexity of WE prices, signaling refinements, complementary monopoly pricing, state-contingent Pareto optima, fixed-point existence, mechanism design/IPVM, mixed equilibrium continuity, verifiable disclosure by a salesman, sequential Cournot, and SEU versus WARP.

1 How to Use This Playbook

1.1 The central exam pattern

The sample exams are broad, but they are not random. Most questions ask you to do one of five things:

1. define an object precisely;
2. characterize an optimizer, demand, supply, price, contract, or equilibrium;
3. prove a monotonicity, existence, welfare, or comparative-statics claim;
4. construct a counterexample or explain why a theorem does not apply;
5. verify an equilibrium using the correct notion of optimality and beliefs.

The four-hour format rewards a clean protocol more than encyclopedic memory. The best answers begin by declaring the mathematical object, then move through feasibility, optimality, consistency, and

conclusion. The most common weak answers jump into algebra without defining the object or forget a boundary/off-path case.

1.2 The universal four-layer answer

For almost every question, write in four layers:

1. **Object.** What is being solved? A consumer problem, firm problem, WE, NE, BNE, WPBE, mechanism, correspondence, or Pareto problem?
2. **Feasibility.** What are the constraints, domains, budget sets, strategy sets, message sets, endowments, or IC/IR conditions?
3. **Optimality/equilibrium.** What is each actor maximizing or minimizing? Which theorem or FOC/KKT/envelope/single-crossing condition applies?
4. **Verification.** Check boundary cases, market clearing, Walras' Law, beliefs, zero profit, incentive compatibility, participation, and uniqueness or nonuniqueness if asked.

1.3 One-hour question protocol

A four-question exam means approximately one hour per question. Use this pacing unless a question is explicitly shorter.

Time	Task
0–5 min	Read all parts. Mark the object in each part: define, compute, prove, counterexample, equilibrium, welfare, diagram. Do not calculate yet.
5–12 min	Write definitions and constraints. If the problem is GE or game theory, define the equilibrium object before solving. If it is optimization, write the feasible set and objective.
12–40 min	Execute the main characterization. Keep notation consistent. If the algebra becomes long, switch to a theorem/proof move rather than grinding blindly.
40–52 min	Verification pass: boundary cases, constraints, beliefs, market clearing, zero profit, off-path behavior, weak versus strict inequalities.
52–60 min	Write the conclusion. State whether the object is unique, nonunique, exists, does not exist, or requires additional assumptions. Add a sentence explaining the economics.

1.4 Universal exam checklist

Before leaving any question, ask:

- Did I define the object requested? Demand correspondence, WE, PO, BNE, WPBE, SE, DRM, etc.
- Did I distinguish function from correspondence?
- Did I state the domain? Nonnegative orthant, compact interval, price simplex, positive prices, finite type set, message set.

- Did I check corners? Zero consumption, zero output, boundary prices, nonparticipation, off-path messages.
- Did I verify feasibility and optimality separately?
- In games, did I specify complete contingent plans, not just observed-path actions?
- In Bayesian/sequential games, did I specify beliefs and consistency where required?
- In GE, did I check all markets or use Walras' Law correctly?
- In mechanism/screening, did I include IC and IR/participation constraints for every type?
- Did I answer the exact comparative-static claim: weakly increasing, strictly increasing, no decrease, no increase, or impossible to say?

2 Prototype Index: Recognize the Question Fast

Prototype	First thing to write	Main tool	Typical trap
Quasilinear demand	Budget exhaustion and reduce to one-dimensional problem	KKT, concavity, value comparison	Forgetting $x_2 \geq 0$ corner
No-Giffen/normality	Compare two optimality inequalities	Revealed-preference addition	Proving too strong a statement
Endowment demand change	Write wealth as $p \cdot \omega$	WARP or SEU FOC	Treating wealth as fixed
Single crossing comparative statics	State SSCP or increasing differences	Topkis/monotone selection	Confusing strict and weak monotonicity
Value function curvature	Express value as supremum of affine functions	Supremum of affine functions is convex	Assuming concavity because objective is maximized
Cost/profit problem	Define technology and feasible set	Cost minimization, homogeneity, Shephard-type logic	Ignoring closedness/existence or zero-output case
WE from excess demand	Derive $z(p)$ and solve $z(p) = 0$ up to normalization	Walras' Law, homogeneity, boundary condition	Solving only one market without justification
WE existence theorem	List conditions on z	continuity, homogeneity, Walras' Law, bounded below, boundary	Missing boundary condition
No-WE example	Pick preferences causing excess demand not to clear	Direct contradiction	Claiming no WE without checking all prices
Edgeworth risk sharing	Draw state-contingent Edgeworth box	MRS equalization or concavity	Missing positivity/interior qualifier

Welfare theorem argument	State feasible allocation and price support	LNS, convexity, supporting prices	Applying theorem when assumptions fail
Strategic-form NE	Write payoff inequalities for each player	Best replies	Listing outcomes rather than strategies
Mixed NE	Make players indifferent over support	Support enumeration	Forgetting off-support inequalities
Weak dominance	Quantify over all opponent strategies	Inequality comparison	Assuming weakly dominated strategies cannot appear in NE
SPNE	Solve every subgame	Backward induction/continuation values	Ignoring strategies after histories not reached
Sequential equilibrium	Strategies + beliefs + sequential rationality + consistency	Bayes rule and limiting fully mixed assessments	Treating beliefs as free on path
BNE	Type-contingent strategy and expected payoff	Interim best response	Forgetting expectation over other types
Signaling WPBE	Actions/messages, beliefs after signals, receiver optimality	Pooling/separating verification	Missing off-path beliefs or zero-profit wage
Verifiable disclosure	List messages each type can send	No-lying constraints and unraveling	Allowing a type to send impossible message
Screening menu	Write IC/IR for every type	Single crossing, binding constraints	Binding the wrong IC constraint
Moral hazard	Choose contract to induce effort	IC, PC, MLRP, FOA	Treating effort as contractible
Mechanism design	Define allocation and transfer functions	Revelation principle, envelope, monotonicity	Forgetting IC utility derivative condition
Auction revenue max	Compute virtual value and allocation rule	Myerson logic	Allocating on value instead of virtual value
Counterexample question	Identify which hypothesis fails	Minimal model construction	Using a counterexample that violates the question's maintained assumptions

3 Universal Definitions You Must Be Able to Write Cleanly

This section gives compact definitions in exam-ready form. Adapt notation to the question.

3.1 Consumer demand and revealed preference

Given prices $p \in \mathbb{R}_{++}^L$, wealth $w > 0$, consumption set $X \subseteq \mathbb{R}_+^L$, and preference relation $\succeq$, the budget set is

$$B(p, w) = \{x \in X : p \cdot x \leq w\}.$$

The Walrasian demand correspondence is

$$d(p, w) = \{x \in B(p, w) : x \succeq y \text{ for all } y \in B(p, w)\}.$$

If preferences are represented by u , the indirect utility function is

$$V(p, w) = \max_{x \in B(p, w)} u(x),$$

when the maximum exists. The expenditure function for utility target $\bar{u}$ is

$$e(p, \bar{u}) = \min_{x \in X} \{p \cdot x : u(x) \geq \bar{u}\}.$$

For a demand function d , WARP for prices p^0, p^1 means: if $x^t = d(p^t, w^t)$ and $p^0 \cdot x^1 \leq p^0 \cdot x^0$, then $p^1 \cdot x^0 > p^1 \cdot x^1$ whenever $x^0 \neq x^1$; adapt the weak/strict inequality to the version stated in the question.

3.2 Competitive/Walrasian equilibrium

For a pure exchange economy with consumers $i = 1, \dots, I$, consumption sets X_i , preferences $\succeq_i$, and endowments ω_i , a Walrasian equilibrium is a price vector p^* and allocation $(x_i^*)_i$ such that:

1. $x_i^* \in B_i(p^*) = \{x_i \in X_i : p^* \cdot x_i \leq p^* \cdot \omega_i\}$ for every i ;
2. x_i^* maximizes $\succeq_i$ on $B_i(p^*)$ for every i ;
3. markets clear: $\sum_i x_i^* = \sum_i \omega_i$.

For a private-ownership production economy, add firms j with production sets Y_j , ownership shares θ_{ij} , profit-maximizing production plans $y_j^* \in \arg \max_{y_j \in Y_j} p^* \cdot y_j$, and consumer wealth

$$w_i = p^* \cdot \omega_i + \sum_j \theta_{ij} p^* \cdot y_j^* + T_i$$

if transfers T_i are present.

3.3 Pareto optimality

An allocation $x = (x_i)_i$ is feasible if $\sum_i x_i \leq \sum_i \omega_i$ in an exchange economy, or if $\sum_i x_i \leq \sum_i \omega_i + \sum_j y_j$ for some feasible production plans $y_j \in Y_j$ in a production economy. A feasible allocation x^* is Pareto optimal if there is no feasible allocation x such that $x_i \succeq_i x_i^*$ for all i and $x_k \succ_k x_k^*$ for at least one k . Strong Pareto optimality usually means strict improvement for at least one person and weak improvement for all others is impossible.

3.4 Strategic-form game and Nash equilibrium

A strategic-form game is a tuple

$$G = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$$

where N is the set of players, S_i is player i 's strategy set, and $u_i : S \rightarrow \mathbb{R}$ is player i 's payoff function on $S = \prod_i S_i$. A strategy profile s^* is a Nash equilibrium if for every player i ,

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \text{for all } s_i \in S_i.$$

A mixed equilibrium replaces S_i by probability distributions over pure strategies and uses expected payoffs.

3.5 Extensive-form strategy, SPNE, and sequential equilibrium

A pure strategy in an extensive-form game is a complete contingent plan: it specifies an action at every information set of the player, including information sets not reached by the strategy profile.

A strategy profile is subgame-perfect Nash equilibrium (SPNE) if its restriction to every subgame is a Nash equilibrium of that subgame. In finite perfect-information games, backward induction is the default method, but when actions are continuous or best replies are not unique, state continuation values and tie-breaking carefully.

A sequential equilibrium is an assessment (σ, μ) where σ is a strategy profile and μ is a belief system satisfying:

1. sequential rationality: at every information set, each player's continuation strategy is optimal given beliefs and continuation strategies;
2. consistency: beliefs are obtained as limits of beliefs generated by completely mixed strategy profiles converging to σ .

For many qualifier answers, Bayes' rule on the equilibrium path plus a clear limiting argument off path is enough.

3.6 Bayesian games, BNE, and wPBE

A Bayesian game specifies players, type sets, common prior, action sets, and payoff functions depending on types and actions. A pure strategy for player i is a function $s_i : T_i \rightarrow A_i$. A Bayesian Nash equilibrium is a strategy profile s^* such that for each player i and type t_i ,

$$\mathbb{E}[u_i(s_i^*(t_i), s_{-i}^*(t_{-i}), t_i, t_{-i}) \mid t_i] \geq \mathbb{E}[u_i(a_i, s_{-i}^*(t_{-i}), t_i, t_{-i}) \mid t_i]$$

for every feasible action a_i .

A weak perfect Bayesian equilibrium in a signaling or disclosure game is an assessment consisting of strategies and beliefs such that strategies are sequentially rational at every information set and beliefs are derived from Bayes' rule whenever possible. Off-path beliefs are unrestricted unless the problem imposes refinements or common-belief restrictions.

3.7 Direct revelation mechanism and IC

In an independent-private-values model, a direct revelation mechanism (DRM) consists of allocation probabilities $q_i(x)$ and transfers $t_i(x)$ for every reported type profile x . For bidder/agent i with true type

x_i , interim utility is

$$U_i(x_i) = \mathbb{E}_{x_{-i}}[x_i q_i(x_i, x_{-i}) - t_i(x_i, x_{-i})].$$

Incentive compatibility means truthful reporting maximizes interim utility:

$$U_i(x_i) \geq \mathbb{E}_{x_{-i}}[x_i q_i(\hat{x}_i, x_{-i}) - t_i(\hat{x}_i, x_{-i})] \quad \text{for all } x_i, \hat{x}_i.$$

For a one-dimensional quasilinear type with differentiability, IC implies the envelope relation

$$U'_i(x_i) = Q_i(x_i),$$

where Q_i is the interim allocation probability, and IC also requires Q_i to be nondecreasing.

4 Demand, Optimization, and Comparative Statics

4.1 Quasilinear demand in two goods

Exam anchors. June 2018 Q1–Q2; June 2020 Q1; August 2016 Q3; June 2017 Q1.

Typical form:

$$u(x_1, x_2) = f(x_1) + x_2, \quad x \in \mathbb{R}_+^2, \quad p \cdot x \leq w.$$

First move. Use local nonsatiation or monotonicity in x_2 to set the budget constraint at equality. Reduce the problem to choosing x_1 over an interval.

If $p = (p_1, p_2) \gg 0$, budget exhaustion gives

$$x_2 = \frac{w - p_1 x_1}{p_2}, \quad 0 \leq x_1 \leq \frac{w}{p_1}.$$

The objective becomes

$$f(x_1) + \frac{w - p_1 x_1}{p_2} =$$

$\frac{w}{p_2} + \left[f(x_1) - \frac{p_1}{p_2} x_1 \right]$. So demand for good 1 solves

$$x_1 \in \arg \max_{0 \leq x \leq w/p_1} \left\{ f(x) - \frac{p_1}{p_2} x \right\}.$$

Then $x_2 = (w - p_1 x_1)/p_2$.

Protocol.

1. State budget exhaustion. If x_2 is the numeraire-good term and u is increasing in x_2 , all wealth is spent.
2. Reduce to one-dimensional maximization.
3. If f is differentiable and concave, solve $f'(x_1) = p_1/p_2$ subject to $0 \leq x_1 \leq w/p_1$.
4. Check corners explicitly: $x_1 = 0$ and $x_1 = w/p_1$.
5. Recover x_2 from the budget.
6. If f is not concave, use global comparisons or correspondence language. Do not write a unique FOC solution unless uniqueness is justified.

Common trap. The most common error is to solve the unconstrained FOC and forget $x_2 \geq 0$. Many exam questions are designed so the unconstrained x_1 may exceed w/p_1 .

4.2 No-Giffen and normality proofs by adding optimality inequalities

Exam anchors. June 2018 Q2; June 2020 Q1; August 2024 Q4; August 2013 Q3.

When asked to prove that demand does not decrease when a price decreases or wealth increases, do not start by differentiating unless the problem grants differentiability and uniqueness. Use two optimality inequalities.

Proof skeleton.

Suppose x^0 is chosen at price/wealth (p^0, w^0) and x^1 is chosen at (p^1, w^1) . Write optimality of x^0 in its feasible set and optimality of x^1 in its feasible set. In quasilinear problems, after reducing to one dimension, for $t = 0, 1$:

$$f(x_1^t) - \alpha^t x_1^t \geq f(x_1^{1-t}) - \alpha^t x_1^{1-t},$$

where $\alpha^t = p_1^t/p_2^t$. Add the two inequalities:

$$(\alpha^0 - \alpha^1)(x_1^1 - x_1^0) \geq 0.$$

If $\alpha^1 < \alpha^0$, conclude $x_1^1 \geq x_1^0$.

For wealth normality at fixed prices in quasilinear problems, compare feasible intervals. If wealth expands the interval, a maximizer at the larger wealth can be chosen at least as large as an old maximizer. But a stronger statement about *every* maximizer may fail when the objective has multiple separated global maxima.

Common trap. Different claims require different strength. “There exists a new maximizer with larger x_1 ” is weaker than “every new maximizer has larger x_1 .” The exams often ask exactly this distinction.

4.3 Kuhn-Tucker protocol for nonnegative consumption

Exam anchors. August 2016 Q3; August 2024 Q4; June 2022 Q2; August 2021 Q4.

For

$$\max_{x \geq 0} U(x) \quad \text{s.t. } p \cdot x \leq w,$$

with U differentiable and locally nonsatiated, the budget binds. Let $\lambda \geq 0$ be the multiplier on $w - p \cdot x \geq 0$. KKT conditions are

$$\frac{\partial U}{\partial x_\ell}(x) - \lambda p_\ell \leq 0, \quad x_\ell \left(\frac{\partial U}{\partial x_\ell}(x) - \lambda p_\ell \right) = 0,$$

for every good ℓ , plus $p \cdot x = w$ and $x \geq 0$.

For subjective expected utility over states,

$$U(x) = \sum_s \pi_s u(x_s),$$

KKT becomes

$$\pi_s u'(x_s) - \lambda p_s \leq 0, \quad x_s [\pi_s u'(x_s) - \lambda p_s] = 0.$$

If $x_s > 0$ and $x_r > 0$, then

$$\frac{\pi_s u'(x_s)}{p_s} = \frac{\pi_r u'(x_r)}{p_r}.$$

Protocol.

1. Establish budget binding by LNS or strict monotonicity.

2. Write KKT with inequality direction carefully.
3. Separate interior and corner cases.
4. Use strict concavity/quasiconcavity to justify sufficiency and uniqueness if available.
5. For state-contingent problems, compare states using ratios π_s/p_s and monotonicity of u' .

If stuck.

If differentiability arguments are messy, use a swap argument: compare $x = (x_1, x_2)$ to the portfolio/state-swapped bundle (x_2, x_1) when prices or probabilities have an ordering. This is exactly the right style for SEU testable-implication questions.

4.4 Indirect utility with endowment choice

Exam anchors. June 2024 Q1.

Form:

$$\max_{\omega \in C} V(p, p \cdot \omega),$$

where C is a compact set of possible endowments.

First move. Recall that indirect utility is increasing in wealth and homogeneous of degree zero in (p, w) under standard assumptions. Since wealth equals $p \cdot \omega$, the consumer chooses the endowment in C that maximizes the value of the endowment at prices p .

If V is strictly increasing in wealth, then the problem is equivalent to

$$\max_{\omega \in C} p \cdot \omega.$$

Thus, if ω^0 solves at p^0 and ω^1 solves at p^1 , the standard revealed-preference inequalities are

$$p^0 \cdot \omega^0 \geq p^0 \cdot \omega^1, \quad p^1 \cdot \omega^1 \geq p^1 \cdot \omega^0.$$

Adding gives

$$(p^1 - p^0) \cdot (\omega^1 - \omega^0) \geq 0.$$

This is the strongest general conclusion. Component-by-component monotonicity need not follow unless the price change has special form and C has suitable lattice/single-crossing structure.

Common trap. Do not treat w as exogenous. The endowment choice affects wealth through $p \cdot \omega$.

4.5 Single crossing, increasing differences, and monotone selection

Exam anchors. August 2018 Q2; June 2023 Q4; August 2014 Q2; June 2013 Q2; August 2023 Q3.

Strict single crossing in (x, t) says, roughly, that a higher t makes higher x relatively more attractive. Increasing differences is the common differentiable/supermodular version:

$$f(x', t') - f(x, t') \geq f(x', t) - f(x, t) \quad \text{for } x' > x, \quad t' > t.$$

Strict increasing differences replaces weak by strict where appropriate.

For $f(x, t) = tg(x) + h(x)$, if g is strictly increasing, then for $x' > x$ and $t' > t$,

$$[f(x', t') - f(x, t')] - [f(x', t) - f(x, t)] = (t' - t)[g(x') - g(x)] > 0.$$

So f has strictly increasing differences, hence SSP.

Protocol.

1. Write the difference-of-differences expression.
2. Show its sign using the primitive monotonicity assumption.
3. Invoke monotone comparative statics only after existence of maximizers is established.
4. State the monotone-selection conclusion correctly: if $t' > t$, then every selection from the higher problem is weakly above every selection from the lower problem under the appropriate theorem; with weaker assumptions, only extremal selections may be monotone.

Common trap. SSP is ordinal; increasing differences is cardinal. If the question asks for SSP, you may prove increasing differences as a sufficient condition, but do not silently assume differentiability.

4.6 Value function curvature

Exam anchors. August 2018 Q2; June 2024 Q1; August 2017 Q4.

If

$$V(t) = \max_{x \in X} \{tg(x) + h(x)\},$$

then V is the pointwise supremum of affine functions of t :

$$t \mapsto tg(x) + h(x).$$

Therefore V is convex in t when the maximum exists. Convexity does not require concavity of h or g .

Proof skeleton.

For t^a, t^b and $\lambda \in [0, 1]$,

$$\begin{aligned} V(\lambda t^a + (1 - \lambda)t^b) &= \sup_x \{[\lambda t^a + (1 - \lambda)t^b]g(x) + h(x)\} \\ &\leq \lambda \sup_x \{t^a g(x) + h(x)\} + (1 - \lambda) \sup_x \{t^b g(x) + h(x)\} \\ &= \lambda V(t^a) + (1 - \lambda)V(t^b). \end{aligned}$$

Common trap. A maximized objective can generate a convex value function in a parameter even if the decision problem itself is a maximization. The curvature is about the parameter, not the choice variable.

4.7 Cost functions and firm problems

Exam anchors. August 2015 Q2; June 2016 Q4; June 2023 Q3; June 2014 Q1; August 2022 Q4.

For a production set Y , output vector q , input vector z , and input prices $w \gg 0$, a cost function is

$$c(q, w) = \min_z \{w \cdot z : (q, -z) \in Y\}.$$

Protocol.

for cost homogeneity under constant returns:

1. Show $c(\lambda q, w) \leq \lambda c(q, w)$ by scaling a feasible input vector for q .

2. Show $c(q, w) \leq (1/\lambda)c(\lambda q, w)$ for $\lambda > 0$ by scaling back a feasible input vector for λq .
3. Handle $\lambda = 0$ separately; zero output should be feasible at zero input under the assumptions or by free disposal.
4. Conclude $c(\lambda q, w) = \lambda c(q, w)$.

For input-demand monotonicity, use revealed-cost minimization. If z^r minimizes at input price r and $z^{r'}$ minimizes at higher price r' , write the two cost inequalities and add them. This typically yields a sign restriction on the input whose price changed.

Common trap. Do not apply differentiability/envelope results when the technology has discrete capital or nonconvexities unless the problem grants differentiability. Use revealed-cost inequalities instead.

5 General Equilibrium, Excess Demand, and Welfare

5.1 Deriving individual and aggregate excess demand

Exam anchors. June 2020 Q3; August 2024 Q1; June 2021 Q1; August 2022 Q1.

For each consumer i :

$$z_i(p) = x_i(p, p \cdot \omega_i) - \omega_i.$$

Aggregate excess demand is

$$z(p) = \sum_i z_i(p) = \sum_i x_i(p, p \cdot \omega_i) - \sum_i \omega_i.$$

Protocol.

1. Normalize prices: either set one price equal to 1, use relative price $r = p_1/p_2$, or impose $\sum_\ell p_\ell = 1$. State the normalization.
2. Solve each consumer's demand at wealth $p \cdot \omega_i$.
3. Subtract endowment to get individual excess demand.
4. Sum to get aggregate $z(p)$.
5. Use Walras' Law to reduce the number of equations if prices are strictly positive.
6. Check whether the solved price vector satisfies the original domain. If the problem requires $p \gg 0$, zero prices are not allowed.

5.2 Excess-demand properties and WE existence

Exam anchors. June 2021 Q1; August 2024 Q1; June 2020 Q3; August 2022 Q1.

A standard existence theorem for aggregate excess demand on $\mathbb{R}_{++}^L$ uses conditions such as:

1. continuity on $\mathbb{R}_{++}^L$;
2. homogeneity of degree zero: $z(\alpha p) = z(p)$ for all $\alpha > 0$;
3. Walras' Law: $p \cdot z(p) = 0$ for all $p \gg 0$;
4. boundedness below: for each good ℓ , $z_\ell(p) > -s$ for some s independent of p ;

5. boundary condition: if a sequence of normalized prices approaches the boundary of the simplex, then excess demand for some good whose price approaches zero becomes arbitrarily large.

First move. When asked whether a given z satisfies these conditions, test them one at a time. Do not infer Walras' Law from homogeneity, and do not infer existence from continuity alone.

Proof skeleton.

for no WE: If $z(p) = (-1, p_1/p_2)$, then no $p \gg 0$ can satisfy $z(p) = 0$ because the first component is always -1 . Even if other properties hold, the absence of a zero is direct.

Common trap. Walrasian equilibrium requires all components of excess demand to be zero if prices are strictly positive. If prices are normalized in the nonnegative simplex and free disposal is allowed, the equilibrium condition may be stated differently; read the problem.

5.3 Walras' Law and missing market clearing

Exam anchors. August 2017 Q1; June 2020 Q4; June 2016 Q1.

If each consumer exhausts the budget and firms maximize profits, aggregate budget balance implies

$$p \cdot z(p) = 0.$$

If $L - 1$ markets clear and $p_L > 0$, then the last market clears:

$$0 = p \cdot z(p) = \sum_{\ell=1}^{L-1} p_{\ell} z_{\ell}(p) + p_L z_L(p) = p_L z_L(p),$$

so $z_L(p) = 0$.

Common trap. You need $p_L > 0$ to divide by p_L . If the missing good has zero price, Walras' Law alone only gives $p_L z_L = 0$.

5.4 Closed graph of equilibrium correspondences

Exam anchors. August 2018 Q3.

If $Q(\omega) = \{\hat{p} : F(\hat{p}, \omega) = 0\}$ and F is continuous, then Q has a closed graph.

Proof skeleton.

Take a convergent sequence $(\hat{p}^n, \omega^n) \rightarrow (\hat{p}, \omega)$ with $\hat{p}^n \in Q(\omega^n)$. Then $F(\hat{p}^n, \omega^n) = 0$ for all n . By continuity,

$$F(\hat{p}, \omega) = \lim_n F(\hat{p}^n, \omega^n) = 0.$$

Therefore $\hat{p} \in Q(\omega)$, proving the graph is closed.

Common trap. Closed graph is not the same as upper hemicontinuity unless compact-valuedness and other conditions are present.

5.5 Edgeworth boxes and Pareto optimality

Exam anchors. August 2015 Q1; August 2018 Q4; August 2023 Q1; June 2022 Q1.

Visual cue. Draw the aggregate endowment as the size of the box. Put consumer 1's origin at southwest and consumer 2's origin at northeast. Mark the endowment. Budget lines have slope $-p_1/p_2$ in consumer

1's coordinates. Pareto optima are tangencies of indifference curves or boundary points where no mutually beneficial move exists.

For differentiable interior PO with two consumers and two goods/states:

$$\frac{\partial U_1 / \partial x_{11}}{\partial U_1 / \partial x_{12}} = \frac{\partial U_2 / \partial x_{21}}{\partial U_2 / \partial x_{22}}.$$

For SEU state-contingent consumption:

$$\frac{\pi_1 u'_i(x_1^i)}{\pi_2 u'_i(x_2^i)}$$

is consumer i 's MRS between state-1 and state-2 consumption.

Protocol.

for risk-sharing PO:

1. Write feasibility state by state: $x_1^1 + x_1^2 = \omega_1$ and $x_2^1 + x_2^2 = \omega_2$.
2. If aggregate endowment is equal across states and consumers share beliefs and are strictly risk averse, full insurance allocations are PO: each person's consumption is constant across states.
3. If aggregate endowment is higher in state 1 than state 2, positive PO allocations give higher consumption in state 1 for each consumer under common beliefs and strict concavity.
4. If positivity is removed, recheck boundary cases. The conclusion may fail because KKT inequalities replace equal MRS conditions.

Common trap. Interior FOCs are not valid at boundary allocations. If the problem says "positive Pareto optimum," use tangency. If it removes positivity, either prove KKT boundary inequalities suffice or give a boundary counterexample.

5.6 First and Second Welfare Theorem moves

Exam anchors. August 2015 Q1; August 2017 Q2; June 2022 Q1; August 2016 Q4.

First Welfare Theorem protocol.

1. Suppose a feasible allocation y weakly improves all consumers and strictly improves one.
2. By optimality at prices p^* , any strictly preferred bundle must be more expensive under LNS/local nonsatiation logic.
3. Sum the budget inequalities across consumers.
4. Contradict feasibility and market clearing.

Second Welfare Theorem protocol.

1. Identify a Pareto optimal allocation x^* .
2. Use convexity of preferences/production and local nonsatiation to support it by a nonzero price vector.
3. Assign transfers so every consumer can afford x_i^* .
4. Verify x_i^* is optimal at the new budget and markets clear.

Common trap. The conclusion of the SFWT may hold in a special economy even when textbook assumptions fail, or assumptions may hold but the requested equilibrium is a quasi-equilibrium rather than a full equilibrium. State exactly what version is being used.

5.7 Robinson Crusoe and representative firm economies

Exam anchors. June 2014 Q1; June 2022 Q1; June 2023 Q3; August 2016 Q4.

First move. Write the firm problem and the consumer budget separately, even with one person. The person owns the firm, but the firm still maximizes profit at prices.

For one firm with production set Y , consumer endowment ω , and ownership of profits, a CE is (x^*, y^*, p^*) such that:

1. $y^* \in \arg \max_{y \in Y} p^* \cdot y$;
2. x^* maximizes $u(x)$ subject to $p^* \cdot x \leq p^* \cdot \omega + p^* \cdot y^*$;
3. $x^* = \omega + y^*$.

Common trap. If the production technology has constant returns, profit maximization may be unbounded or have zero profit only at particular prices. Supply may be a correspondence, not a function.

6 Strategic-Form, Extensive-Form, and Repeated Games

6.1 Writing strategy sets

Exam anchors. June 2023 Q1; August 2024 Q2; June 2018 Q3; August 2021 Q1; June 2016 Q2.

A strategy is not an action. A strategy is a complete plan.

Protocol.

1. List every information set of the player.
2. For each information set, list feasible actions.
3. A pure strategy is a function assigning one feasible action at each information set.
4. If the game has types, a pure strategy maps each type to an action or continuation strategy.
5. If the game has histories, a strategy maps each possible history at which the player moves to an action.

Examples:

- In a two-type entry/R&D game, a firm strategy is a pair such as (Y, N) : enter if strong, not enter if weak.
- In a signaling game, a sender strategy maps each type to a signal; a receiver strategy maps each observed signal to an action.
- In a two-period owner-manager game, the owner's strategy specifies whom to choose at date 1 and whom to choose at date 2 after every date-1 history.
- In an OLG repeated game, a player who lives two dates specifies an action when young for every public history and an action when old for every public history after having been young.

Common trap. When asked for the strategy set, do not answer with equilibrium-path actions only. Include off-path histories.

6.2 Pure-strategy Nash equilibrium

Exam anchors. June 2019 Q1; August 2016 Q1; June 2024 Q2; August 2024 Q2.

Protocol.

1. For each player, compute best replies to each opponent action or strategy.
2. Intersect best-reply correspondences.
3. If actions are continuous, solve FOCs only after concavity and domain restrictions are checked.
4. Verify each candidate by writing payoff inequalities.

For continuous Cournot-type problems, best response usually solves

$$\max_{q_i \geq 0} [P(q_i + q_j) - k_i]q_i.$$

Use concavity of profit or direct comparisons to justify the FOC. Check $q_i = 0$ and any capacity/price-zero boundary.

6.3 Mixed-strategy Nash equilibrium

Exam anchors. June 2018 Q4; June 2024 Q2.

Protocol.

1. Guess supports.
2. Make each player indifferent among all pure strategies in the support.
3. Ensure off-support pure strategies do not give higher payoff.
4. Check probability constraints.
5. Repeat support cases systematically.

Common trap. At parameter values with symmetry or degeneracy, the equilibrium correspondence may expand. Do not assume continuity of equilibrium selections. For continuity questions, distinguish upper hemicontinuity, lower hemicontinuity, and convergence of equilibrium sets.

6.4 Weak dominance and weakly dominated strategies

Exam anchors. August 2024 Q2; June 2024 Q2.

A strategy s'_i weakly dominates s_i if

$$u_i(s'_i, s_{-i}) \geq u_i(s_i, s_{-i}) \quad \text{for all } s_{-i},$$

and strict inequality holds for at least one s_{-i} .

Protocol.

1. Quantify over all opponent strategies, not just equilibrium candidates.
2. For weak dominance, find one strict case and no negative cases.

3. To show no weakly dominant strategy exists, for each candidate construct an opponent strategy against which another strategy does better.
4. Remember: a weakly dominated strategy can be used in a Nash equilibrium when the weak domination is not strict at the equilibrium opponent profile.

Common trap. Never write “weakly dominated strategies cannot appear in equilibrium.” That is false. Strictly dominated strategies cannot appear in equilibrium under standard assumptions; weakly dominated strategies can.

6.5 SPNE by continuation values

Exam anchors. June 2023 Q1; August 2024 Q2; August 2015 Q4; June 2016 Q2.

Protocol.

1. Identify all proper subgames.
2. Solve terminal subgames first. If actions are continuous, characterize the continuation payoff function.
3. Move backward. At each earlier node, choose an action maximizing current payoff plus discounted continuation value.
4. Record strategies after all histories, including histories not reached.
5. Verify that each subgame restriction is a Nash equilibrium.

In owner-manager finite-horizon problems, date-2 managers have no future discipline, so a chosen manager typically expropriates all available profit at date 2. Date-1 behavior can be disciplined only by date-2 selection promises, bounded by the value of being selected later.

Common trap. In a finite horizon, threats that are not optimal in the continuation subgame cannot support SPNE, though they may support Nash equilibrium.

6.6 Sequential equilibrium

Exam anchors. June 2023 Q2; June 2018 Q4; June 2017 Q3; June 2013 Q4.

Protocol.

1. Write the assessment (σ, μ) .
2. At each information set, compute optimal actions as a function of beliefs.
3. Use Bayes' rule to pin down beliefs at reached information sets.
4. For unreached information sets, specify beliefs that can be obtained as limits of fully mixed perturbations.
5. Verify sequential rationality at every information set.
6. Verify consistency or explain the perturbation sequence.

For a receiver information set with top-node belief α , compute expected payoff of each action as a function of α . This partitions $[0, 1]$ into belief regions. Sequential rationality then says the receiver action/mixing must lie in the argmax correspondence for that α .

Common trap. A strategy profile can be a Nash equilibrium but not a sequential equilibrium because it relies on noncredible behavior at an unreached information set.

6.7 Repeated and overlapping-generation games

Exam anchors. June 2019 Q1; August 2021 Q1; August 2013 Q2.

Protocol.

for finite repetition:

1. Solve the stage-game Nash equilibria and payoff set.
2. In the last period, play a stage-game Nash equilibrium in every subgame.
3. Use continuation payoff differences from later periods to support earlier non-stage-NE behavior if possible.
4. For $T = 2, 3, 4$, construct explicit histories and punishments. Do not just cite a folk theorem.

Protocol.

for OLG games:

1. State a player's strategy as a rule for young and old actions after each public history.
2. To support cooperation by the young, use the threat that future partners punish when the player is old.
3. Check incentives separately for young and old players. Old players often have no future, so they may defect unless current payoff incentives favor cooperation.
4. For SPNE, check every public history. Punishments must themselves be credible continuation equilibria.

Common trap. A strategy profile that supports an outcome as Nash may fail SPNE because a punishment after an off-path history is not optimal for future players.

6.8 Game existence questions

Exam anchors. August 2021 Q1; August 2023 Q2; June 2020 Q4; August 2022 Q4.

To invoke Debreu-Fan-Glicksberg or Kakutani-style existence, check:

1. compact, convex strategy sets;
2. nonempty strategy sets;
3. payoff continuity in all strategies;
4. quasiconcavity in own strategy, or best-reply correspondence nonempty, convex-valued, and upper hemicontinuous.

Common trap. Infinite histories can make pure-strategy spaces noncompact or not obviously compact in the relevant topology. Do not invoke a finite-player compact-convex theorem without checking the strategy space.

7 Bayesian Games, Signaling, and Verifiable Disclosure

7.1 Bayesian Nash equilibrium

Exam anchors. August 2018 Q1; June 2016 Q3; June 2021 Q4; June 2022 Q4; August 2023 Q4.

Protocol.

1. Define type sets and prior.
2. Define each player's strategy as a type-contingent function.
3. For each type of each player, compute expected payoff from each action, integrating over other players' types conditional on own type.
4. Write best-response inequalities type by type.
5. Solve the system of inequalities.
6. Compare equilibria by likelihood of play if asked; compute probability of each strategy profile under the prior.

Common trap. In a Bayesian game, a type's deviation changes only that type's action, not the whole type-contingent strategy unless the question asks for a strategy deviation.

7.2 Incomplete-information Cournot

Exam anchors. June 2021 Q4; June 2022 Q4; August 2023 Q4.

Protocol.

1. A strategy for the informed firm maps each cost type to output. A strategy for the uninformed firm is a single output, unless it observes a message before choosing.
2. Derive best replies. For linear demand $P(Q) = A - Q$, interior Cournot FOC for firm i is

$$A - q_i - \mathbb{E}[q_j \mid I_i] - k_i - q_i = 0,$$

with the expectation determined by information.

3. Use symmetry or mean-cost assumptions when provided. Several exams make expected cost equal to the known rival cost, simplifying the uninformed firm's output.
4. Check positivity assumptions before relying on interior formulas.
5. In disclosure extensions, solve continuation games after each message using posterior beliefs.

Common trap. The uninformed firm's best response depends on expected rival output, not expected rival cost directly, unless output is linear in cost and you have already derived that relationship.

7.3 Signaling games and WPBE

Exam anchors. June 2018 Q3; June 2017 Q2; August 2017 Q3; August 2022 Q3; August 2023 Q3; August 2024 Q3.

Protocol.

for a signaling WPBE:

1. Sender strategy: map each type to a signal/education/training level.
2. Receiver strategy: map each observed signal to an action/wage/quantity.
3. Beliefs: after each signal, give posterior probabilities over types.
4. Receiver optimality: receiver's action maximizes expected payoff given beliefs.
5. Sender optimality: each type prefers its assigned signal and resulting receiver action to every alternative signal and continuation action.
6. Consistency: on-path beliefs follow Bayes' rule. Off-path beliefs are free unless restricted.

In competitive labor-market signaling, zero profit usually implies

$$w(e) = \mathbb{E}[\theta \mid e].$$

In a separating equilibrium, different types choose different education levels, so wages equal true productivity. In a pooling equilibrium, all pooling types receive the average productivity conditional on the pool.

7.4 Separating equilibrium construction

First move. Normalize the lowest type's signal to zero whenever education/training is costly and unproductive. Then recursively choose higher signals just high enough to deter the next lower type from mimicking.

For two types L, H with wage equal to productivity in separation and cost $c(e, \theta)$ lower for high type, candidate separation often has

$$e_L = 0, \quad w_L = \theta_L, \quad w_H = \theta_H,$$

and e_H satisfying two IC inequalities:

$$\theta_H - c(e_H, H) \geq \theta_L - c(0, H),$$

$$\theta_L - c(0, L) \geq \theta_H - c(e_H, L).$$

The interval of separating education levels is where both inequalities hold. The least-cost separating equilibrium chooses the smallest e_H satisfying the low type's no-mimic constraint.

Visual cue. In (e, w) space, draw indifference curves for low and high types through their intended contracts. Single crossing means the high type has flatter indifference curves if education is less costly for high type. Separation places the high type northeast of the low type just far enough that the low type does not mimic.

Common trap. The high type's IC may be slack at the least-cost separating equilibrium; the low type's no-mimic constraint is usually binding. But verify from primitives.

7.5 Pooling and partial pooling

Exam anchors. August 2017 Q3; August 2024 Q3; June 2018 Q3.

Protocol.

1. Identify the pool and compute the posterior conditional on the pooling signal.

2. Use zero profit or receiver optimality to determine the pooling wage/action.
3. Verify every type in the pool prefers the pooling signal to deviations.
4. Specify off-path beliefs that make deviations unattractive, if possible.
5. For partial pooling, partition type space and compute average productivity within each cell.

Common trap. A pooling equilibrium may exist under weak PBE because off-path beliefs can punish deviations. It may be eliminated by refinements such as intuitive criterion/D1 when only high types would benefit from a deviation.

7.6 Verifiable disclosure and no-lying games

Exam anchors. June 2024 Q3; June 2021 Q4; June 2022 Q4.

In verifiable disclosure, a type can reveal truth or send a vague message, but cannot lie. Message sets are type-dependent. For example, type L can send m_L or m_0 , type H can send m_H or m_0 .

Protocol.

1. List each type's feasible messages. This is the main constraint.
2. For every message, compute the receiver's posterior using Bayes' rule when on path.
3. Compute receiver demand/action as a function of posterior expected quality/cost.
4. Check whether a type wants to deviate to a more revealing or less revealing message.
5. Look for unraveling: if high type can distinguish itself and benefits from doing so, pooling on vague messages may fail.
6. In no-communication comparisons, compute ex ante payoffs under prior beliefs and compare to each disclosure equilibrium.

Common trap. Do not assign a posterior putting positive probability on a type that could not have sent the observed message. Feasible message sets restrict beliefs.

8 Screening, Adverse Selection, Insurance, and Credit

8.1 The IC/IR menu template

Exam anchors. June 2021 Q3; June 2013 Q5; June 2014 Q3; June 2020 Q2; June 2019 Q4.

A principal offers contracts (q_i, T_i) or (x_i, t_i) to types $i \in \{L, H\}$.

Protocol.

1. Write type i 's utility from contract j , $U_i(c_j)$.
2. IC constraints:

$$U_i(c_i) \geq U_i(c_j) \quad \text{for all } i, j.$$

3. IR/participation constraints:

$$U_i(c_i) \geq \bar{U}_i.$$

4. Principal objective: expected profit, usually $\sum_i \lambda_i \Pi(c_i, i)$.
5. Use single crossing to infer monotonicity of allocation: higher type gets weakly higher quality, coverage, warranty, or output.
6. Identify likely binding constraints. Standard two-type monopoly screening with high willingness to pay: low-type IR and high-type IC bind.
7. Solve the reduced system and verify all omitted constraints.

Common trap. Binding constraints are conclusions, not assumptions. State why the usual constraints bind, usually via rent extraction and single crossing.

8.2 Insurance screening

Exam anchors. June 2014 Q3; June 2020 Q2; August 2013 Q1; June 2021 Q3.

Insurance contracts often have coverage x and transfer/premium t . With loss L and loss probability p , utility can be written

$$U(x, t, p) = pu(w - L + x - t) + (1 - p)u(w - t).$$

The firm's expected profit from a type- p consumer is often

$$t - px.$$

Protocol.

for competitive adverse selection:

1. Draw (x, t) space with x horizontal and t vertical.
2. Zero-profit line for type p is $t = px$.
3. Higher risk has steeper zero-profit line.
4. Indifference curves reflect risk aversion; full insurance often lies at $x = L$.
5. In separating candidates, high-risk types usually receive full insurance and low-risk types receive partial insurance distorted downward to prevent high-risk mimicking.
6. Check whether a pooling or cross-subsidizing deviation by a competitor destroys the candidate. Existence can fail in Rothschild-Stiglitz-style models.

Common trap. In competitive screening, zero profit is not just a feature of equilibrium; profitable deviations by firms determine whether an apparent separating menu actually exists.

8.3 Credit screening with collateral

Exam anchors. June 2019 Q4.

Loan contract (r, x) : interest r , collateral x . Borrower type is insolvency probability q . Borrower payoff:

$$1 - r(1 - q) - qx.$$

Lender payoff:

$$r(1 - q) - qbx,$$

where $0 < b < 1$ captures inefficient collateral realization.

Protocol.

1. Draw x horizontally and r vertically.
2. Zero-profit lines solve $r(1 - q) - qbx = 0$, so

$$r = \frac{qb}{1 - q}x.$$

3. Higher-risk borrowers have steeper zero-profit lines.
4. Borrower indifference lines solve $1 - r(1 - q) - qx = \text{constant}$, so

$$r = \frac{1 - \text{constant} - qx}{1 - q}.$$

5. Use single crossing: collateral is more costly for high-risk borrowers because they lose it more often. Low-risk borrowers can signal by accepting more collateral.
6. Find the unique separating candidate by making high risk receive its full-information zero-collateral contract and low risk just unattractive to high risk.
7. Test for profitable deviations to determine existence.

Common trap. Collateral is inefficient for lenders ($b < 1$), so it is used for screening, not because it is technologically productive.

8.4 Monotone menus from single crossing

Exam anchors. June 2023 Q4; June 2021 Q3; June 2014 Q3; August 2014 Q2.

Suppose $u(q, T, \theta) = f(q, \theta) - T$ and f has strictly increasing differences in (q, θ) . If type $\theta_j > \theta_i$ chooses (q_j, T_j) and type θ_i chooses (q_i, T_i) from a menu, IC gives

$$f(q_i, \theta_i) - T_i \geq f(q_j, \theta_i) - T_j,$$

$$f(q_j, \theta_j) - T_j \geq f(q_i, \theta_j) - T_i.$$

Adding:

$$f(q_j, \theta_j) - f(q_i, \theta_j) \geq f(q_j, \theta_i) - f(q_i, \theta_i).$$

Strict increasing differences imply this is compatible with $\theta_j > \theta_i$ only if $q_j \geq q_i$.

Common trap. Monotonicity of quantity does not automatically imply monotonicity of transfers unless preferences and IC further restrict transfers. Often both move together in the exam's partial order, but prove the requested ordering.

9 Moral Hazard and Principal-Agent Problems

9.1 Baseline moral hazard setup

Exam anchors. June 2019 Q3; August 2021 Q2; June 2022 Q3; June 2015 Q3; June 2014 Q4.

A principal offers a wage schedule $w(x)$ or repayments contingent on observable output/return x . Effort a is chosen privately. The agent's expected utility is

$$\mathbb{E}[u(w(x)) \mid a] - \psi(a).$$

The principal's expected payoff is

$$\mathbb{E}[x - w(x) \mid a]$$

or the lender/owner-specific equivalent.

Protocol.

1. State whether effort is observable. If observable, no IC constraint is needed; if unobservable, effort must be induced.
2. For target effort a , write the participation constraint:

$$\mathbb{E}[u(w(x)) \mid a] - \psi(a) \geq \bar{u}.$$

3. Write the incentive constraint:

$$\mathbb{E}[u(w(x)) \mid a] - \psi(a) \geq \mathbb{E}[u(w(x)) \mid \hat{a}] - \psi(\hat{a}) \quad \text{for all } \hat{a}.$$

4. If effort is continuous and FOA is valid, replace global IC by the first-order condition:

$$\int u(w(x)) f_a(x \mid a) dx = \psi'(a).$$

5. Solve cost minimization for each target effort.
6. Choose effort to maximize principal payoff net of minimized cost.

Common trap. The contract cannot condition directly on effort unless effort is observable or monitoring makes it observable.

9.2 MLRP, FOA, and wage monotonicity

Exam anchors. June 2019 Q3; June 2014 Q4.

The monotone likelihood ratio property (MLRP) says the likelihood ratio $f_a(x \mid a)/f(x \mid a)$ or $f(x \mid a')/f(x \mid a)$ is increasing in x for higher effort. It implies high output is stronger evidence of high effort.

The first-order approach (FOA) is valid under conditions such as MLRP plus convexity of the distribution function condition (CDFC), ensuring the agent's FOC is sufficient for global incentive compatibility.

Protocol.

1. Compute the likelihood ratio or derivative of log density.
2. Check whether it is increasing in output.
3. Check CDFC/concavity conditions if asked whether FOA is valid.
4. In the cost-minimization Lagrangian, the wage/utility assigned to output x depends on a multiplier times the likelihood ratio.
5. If MLRP holds, the optimal wage schedule is increasing in output.

Common trap. MLRP alone may not guarantee FOA validity. State the additional condition if needed.

9.3 Cost-minimizing compensation in utility terms

The cleanest way to solve moral hazard contracts is often to choose promised utility levels $v(x) = u(w(x))$ rather than wages.

For target effort a with density $f(x | a)$, solve

$$\min_{v(\cdot)} \int h(v(x)) f(x | a) dx$$

subject to

$$\int v(x) f(x | a) dx - \psi(a) = \bar{u},$$

$$\int v(x) f_a(x | a) dx = \psi'(a),$$

where $h = u^{-1}$.

FOC:

$$h'(v(x)) = \lambda + \mu \frac{f_a(x | a)}{f(x | a)}.$$

If h' is increasing and the likelihood ratio is increasing, then $v(x)$ and hence $w(x)$ are increasing in x .

Common trap. Do not lose the participation constraint. It often binds because the principal can reduce all wages until the agent reaches reservation utility.

9.4 Monitoring and informativeness

Exam anchors. June 2015 Q3; June 2019 Q3; June 2022 Q3.

A signal y should enter compensation if it provides information about effort beyond output x . In Holmstrom's informativeness principle, include y if the likelihood ratio conditional on (x, y) is not a function of x alone.

Protocol.

1. Compare the likelihood ratio with and without the signal.
2. If y changes the likelihood ratio relevant for effort incentives, it is valuable for contracting.
3. The value of monitoring equals the principal's payoff under observable/monitored effort minus payoff under output-only contracting, net of monitoring cost.

Common trap. Correlation between x and y is not enough. The signal must be informative about effort conditional on output.

10 Mechanism Design and Auctions

10.1 Direct revelation mechanism

Exam anchors. June 2024 Q4; August 2015 Q3; June 2023 Q4.

For bidders $i = 1, \dots, I$ with type spaces X_i , a DRM is a pair of functions:

$$q_i : X \rightarrow [0, 1], \quad t_i : X \rightarrow \mathbb{R},$$

where $q_i(x)$ is bidder i 's probability of receiving the object and $t_i(x)$ is bidder i 's transfer. Feasibility requires

$$\sum_i q_i(x) \leq 1 \quad \text{for all } x.$$

In a single-agent mechanism, feasibility is simply $q(x) \in [0, 1]$.

10.2 IC utility characterization

For one agent with type $x \in [a, b]$, interim allocation $Q(x)$, transfer $T(x)$, and utility

$$U(x) = xQ(x) - T(x),$$

IC is equivalent to:

1. Q is nondecreasing;
2. $U(x) = U(a) + \int_a^x Q(s)ds$;
3. participation requires $U(x) \geq 0$, usually enforced by $U(a) \geq 0$ when $Q \geq 0$.

Given U and $Q = U'$, the transfer is

$$T(x) = xQ(x) - U(x).$$

Protocol.

for “does this utility correspond to a DRM?”:

1. Check U is absolutely continuous.
2. Set $Q(x) = U'(x)$ where differentiable.
3. Check $0 \leq Q(x) \leq 1$.
4. Check Q is nondecreasing.
5. Check participation $U(x) \geq 0$.
6. Define $T(x) = xQ(x) - U(x)$ and verify if needed.

Common trap. A convex-looking utility is not enough. The derivative must be a feasible monotone allocation probability.

10.3 Revenue-maximizing DRM and virtual values

Exam anchors. June 2024 Q4.

For bidder i with value x_i and distribution F_i with density f_i , the virtual value is

$$\phi_i(x_i) = x_i - \frac{1 - F_i(x_i)}{f_i(x_i)}.$$

The revenue-maximizing allocation gives the object to the bidder with the highest nonnegative virtual value, subject to monotonicity/ironing if virtual values are not increasing.

For a single bidder, sell if $\phi(x) \geq 0$ and charge the threshold price. Example: if $F(x) = x^2$ on $[0, 1]$, then $f(x) = 2x$ and

$$\phi(x) = x - \frac{1 - x^2}{2x} = \frac{3x^2 - 1}{2x}.$$

The reserve solves $\phi(r) = 0$, so $r = 1/\sqrt{3}$.

Common trap. Surplus maximization allocates to the highest value; revenue maximization allocates to the highest nonnegative virtual value. They differ when reserves or asymmetric distributions matter.

10.4 Revenue equivalence

Exam anchors. August 2015 Q3.

If two IC mechanisms have the same interim allocation probability $Q_i(x_i)$ for every type, then their interim utilities differ by a constant independent of type. From the envelope formula:

$$U_i(x_i) = U_i(a_i) + \int_{a_i}^{x_i} Q_i(s) ds.$$

So if $Q_i = Q'_i$ for all x_i , then

$$U'_i(x_i) - U_i(x_i) = U'_i(a_i) - U_i(a_i),$$

and transfers differ correspondingly by the same utility normalization.

Common trap. Revenue equivalence compares mechanisms with the same allocation rule and the same utility normalization for the lowest type. Without the normalization, utilities can differ by a constant.

11 High-Yield Proof Moves

11.1 Revealed-preference addition

Use when two choices are optimal under two environments. Write two optimality inequalities and add them. This proves monotonicity, no-Giffen results, input-demand laws, and endowment-choice price inequalities.

Generic form:

$$f(x^0, a^0) \geq f(x^1, a^0), \quad f(x^1, a^1) \geq f(x^0, a^1).$$

Add and simplify. The parameter terms usually determine the sign of $x^1 - x^0$.

11.2 Contradiction with feasibility

Use in welfare theorem proofs and market-clearing proofs. Suppose an improvement exists, use consumer optimality to show it costs strictly more, sum across consumers, contradict aggregate resources.

11.3 Boundary/corner reduction

Use when FOCs fail. Separate cases:

- zero consumption/output/effort;

- price at boundary of simplex;
- education/training equal to zero;
- no trade/no insurance/no participation;
- type pooling versus separating.

State what inequality replaces equality at the boundary.

11.4 Envelope move

Use for indirect utility, value functions, IC mechanisms, and dynamic problems. If $V(t) = \max_x f(x, t)$ and $x(t)$ is optimal, then under differentiability,

$$V'(t) = \frac{\partial f}{\partial t}(x(t), t).$$

But if differentiability is not given, use convexity of value functions or one-sided inequalities.

11.5 Single-crossing monotonicity

Use whenever types choose from a menu and higher types value higher allocations more. Write IC constraints for two types, add them, and invoke increasing differences to rule out nonmonotone allocations.

11.6 Bayes consistency move

Use in WPBE/SE. On path:

$$\mu(t \mid m) = \frac{\pi(t)\mathbf{1}\{s(t) = m\}}{\sum_{t'} \pi(t')\mathbf{1}\{s(t') = m\}}.$$

Off path, beliefs are free in wPBE unless refinements or feasibility of messages restrict them.

11.7 Zero-profit move

Use in competitive labor, insurance, credit, and Bertrand screening. Competition forces the contract/wage offered to a type or pool to yield zero expected profit. Compute the break-even transfer/wage conditional on the posterior type distribution.

12 Counterexample Construction

Many exam questions ask: “Does this follow? If so prove it; if not give a counterexample.” Counterexamples should be minimal and should violate only the target conclusion, not the maintained assumptions.

12.1 Counterexample protocol

1. Identify the conclusion to refute.
2. List the assumptions that must remain true.

3. Choose the smallest environment: usually two goods, two consumers, two actions, or two types.
4. Make the failure visible: a flat segment, nonconvexity, boundary, multiple maximizers, weak inequality, or missing compactness.
5. Verify assumptions one by one.
6. Show the conclusion fails directly.

12.2 Common counterexample templates

Target false claim	Template
Uniqueness of demand	Use perfect substitutes, Leontief kink, or flat objective segment.
Stronger normality of every selection	Create two separated maximizers that remain optimal after wealth changes.
WE price uniqueness	Use symmetric Cobb-Douglas or homothetic exchange where relative price unique but normalization not, or redundant goods.
WE allocation uniqueness	Use non-strict convexity/quasiconcavity.
FWT/SFWT assumptions	Remove local nonsatiation, convexity, or positive prices.
SPNE equals NE	Use noncredible threats in an unreached subgame.
Weakly dominated strategies absent from NE	Use a payoff tie at the equilibrium opponent strategy.
FOC sufficiency	Use nonconcave objective.
Continuity of equilibrium selections	Use a parameter value where multiple equilibria appear.

Common trap. A counterexample that violates the maintained assumptions earns little credit. If the problem assumes LNS and strict quasiconcavity, your counterexample must satisfy both.

13 Visual Fallbacks Embedded in the Playbook

A separate visual curriculum can expand these, but during the exam each diagram should serve a proof.

Diagram	What to label	Use it for
Budget/WARP	x^0, x^1 , two budget lines, endowment ω , relative price change	Revealed preference, endowment-income effects, SEU tests
Quasilinear demand	One-dimensional objective $f(x) - \alpha x$, feasible interval	Corners, no-Giffen proofs
Edgeworth box	Endowment, budget line, contract curve, full-insurance line	GE, welfare, risk sharing
Excess demand on simplex	Price boundary, zero of z , direction of excess demand	Existence, nonexistence, convexity of WE prices
Best-response graph	BR_1, BR_2 , intersections, discontinuities	Cournot, pricing games, NE existence

Game tree	Information sets, beliefs, continuation payoffs	SPNE, SE, WPBE
Signaling/screening graph	Education/quantity horizontal, wage/transfer vertical, indifference curves	Separating/pooling, IC constraints, least-cost separation
Insurance/credit graph	Coverage/collateral horizontal, premium/interest vertical, zero-profit lines	Competitive screening and deviations
Moral hazard wage graph	Output on horizontal, wage/utility on vertical, likelihood ratio slope	MLRP, monotone wages, signal informativeness

14 Exam-Day Triage by Topic

14.1 If the question is general equilibrium

Write:

1. consumers, endowments, budget sets;
2. firm problems if production exists;
3. definition of WE/CE;
4. individual demand/supply;
5. market clearing or excess demand;
6. Walras' Law if using $L - 1$ markets;
7. welfare/existence conditions if asked.

Do not leave without checking price normalization and whether prices must be strictly positive.

14.2 If the question is a game

Write:

1. players;
2. strategy sets as complete contingent plans;
3. payoff functions over strategy profiles;
4. equilibrium definition requested;
5. best replies or sequential rationality;
6. beliefs if incomplete information or imperfect information;
7. off-path histories/messages.

Do not write only an outcome if the question asks for strategies.

14.3 If the question is screening or mechanism design

Write:

1. types and probabilities;
2. contract/menu/mechanism variables;
3. type utility from every contract;
4. IC constraints;
5. IR constraints;
6. principal objective;
7. single-crossing/monotonicity result;
8. binding constraints and verification.

Do not bind constraints before explaining why.

14.4 If the question is moral hazard

Write:

1. observables and unobservables;
2. contract space;
3. target effort;
4. participation constraint;
5. incentive constraint;
6. cost minimization for target effort;
7. effort choice by principal;
8. MLRP/FOA/informativeness checks.

Do not condition wages on effort unless effort is observable.

14.5 If the question is decision theory/uncertainty

Write:

1. state-contingent budget;
2. beliefs and vN-M utility;
3. KKT or swap/revealed-preference argument;
4. risk aversion comparison, if relevant;
5. certainty equivalent or mean-variance expression, if relevant.

Do not assume concavity if the problem says the vN-M utility is increasing but not necessarily concave.

15 Common Traps and Fast Corrections

Trap	Fast correction
Solving FOC without checking concavity	State concavity/strict quasiconcavity or verify global optimum.
Forgetting nonnegativity	Add KKT inequalities and corner checks.
Treating a correspondence as a function	Use set-valued notation and selection language.
Ignoring price normalization	State whether prices are in $\mathbb{R}_{++}^L$, simplex, or one good is numeraire.
Using Walras' Law to clear a zero-price market	Need strictly positive price for the missing market.
Listing actions instead of strategies	Include full contingent plans.
Calling a NE subgame perfect without checking subgames	Verify every subgame.
Ignoring off-path beliefs in WPBE/SE	Specify beliefs and receiver optimality.
Assuming weakly dominated strategies cannot be in NE	Only strictly dominated strategies are excluded generally.
Binding the wrong IC constraint	Use single crossing and verify all omitted IC constraints.
Assuming pooling impossible in signaling	Weak PBE can support pooling with off-path beliefs.
Forgetting zero profit under competition	Compute wage/contract from posterior expected productivity/cost.
Treating moral-hazard effort as contractible	Add IC; effort is chosen by agent after contract.
Using FOA without validity conditions	Check MLRP/CDFC or explain global IC.
Using virtual values incorrectly	Revenue uses virtual values, surplus uses values.
Claiming theorem applies without hypotheses	List compactness, convexity, continuity, LNS, monotonicity, etc.

16 Practice Postmortem Template

After every timed problem, fill this out in two minutes.

Field	Question to answer
Exam/question	Which exam and question?
Prototype	Which prototype from the index?
First move	What should I have written in the first five minutes?
Object	Did I define the correct object?
Key theorem/tool	What unlocked the problem?
Diagram	What picture or logic map should I have drawn?

Boundary/off-path	What corner, boundary, or off-path case mattered?
Mistake type	Algebra, definition, theorem hypothesis, time, diagram, or interpretation?
Redo trigger	What condition tells me to redo this problem?
One-sentence lesson	What do I want future-me to remember?

17 Appendix A: Exam Archetype Map

This map is not a solution guide. It is a navigation aid for selecting practice problems.

Exam	High-yield archetypes
June 2013	Owner-manager SPNE and non-SPNE NE; increasing differences and MLRP; Cournot rationalizability; sequential equilibrium in Selten's horse variant; monopoly screening.
August 2013	Monopoly insurance and risk aversion; OLG prisoners' dilemma; net-buyer demand after price change; signaling versus screening; binary-output moral hazard with limited liability.
June 2014	Robinson Crusoe WE; expenditure/minimum-wealth representation; insurance screening with SSCP; moral hazard and MLRP.
August 2014	Supporting any strictly positive bundle with prices/endowment; nonmarket good and indirect utility concavity; demand and excess demand theorem; owner-manager Cournot contracts.
June 2015	LNS budget implications; more concave utility and certainty equivalents; moral hazard monitoring; Cournot existence with discontinuous best replies.
August 2015	Edgeworth WE and SWFT transfers; CRS cost homogeneity; direct mechanisms/revenue equivalence; finite sequential-move SPNE characterization.
June 2016	Exchange with nontraded good and Pareto rankings; parent-child SPNE; Bayesian coordination; input demand with discrete capital.
August 2016	Advertising NE, Pareto optimum, and core; public project/risk-sharing game; KKT demand and wealth expansion path; Robinson economy with nonconvex production.
June 2017	Expenditure and normality; market signaling WPBE; sequential equilibria; public good bidding with unknown cost and risk aversion.
August 2017	AD market clearing by Walras' Law; local nonsatiation and FWT logic; partial-pooling signaling; dynamic demand complementarity.
June 2018	Quasilinear demand and CE; no-Giffen proof; signaling WPBE; NE and sequential equilibrium.
August 2018	Bayesian R&D entry; single crossing and value function convexity; closed graph of equilibrium prices; uncertainty Edgeworth box and timing of trade.
June 2019	Finite repeated games; advertising NE/PO/core; moral hazard MLRP/FOA/informativeness; credit-market screening.
June 2020	Quasilinear demand normality; competitive insurance adverse selection; exchange excess demand and WE existence; abstract economy/auctioneer and strategic form.

June 2021	Excess-demand properties and nonexistence; flexibility value; warranty screening; Bayesian Cournot and disclosure.
August 2021	OLG repeated prisoners' dilemma; two-output moral hazard; continuous-type signaling; quadratic utility portfolio choice.
June 2022	Robinson Crusoe WE and welfare theorem edge cases; SEU testable implications; moral hazard lending and monitoring; Bayesian Cournot disclosure.
August 2022	Excess-demand WARP and convexity of WE prices; more risk-averse consumer state demand; least-cost separating signaling; complementary monopoly price game.
August 2023	State-contingent Pareto optimality; fixed-point existence; finite-type Spence signaling recursion; Bayesian Cournot.
June 2023	Owner-manager SPNE; sequential equilibrium characterization; Robinson CE; single crossing and mechanism-design menu monotonicity.
June 2024	Indirect utility and endowment choice; mixed NE and equilibrium correspondence continuity; verifiable disclosure WPBE; IPV direct mechanisms and optimal auctions.
August 2024	Excess demand, WE existence, and nonexistence; sequential Cournot, dominance, and SPNE; signaling/screening threshold equilibria; WARP versus SEU testable implications.

18 Appendix B: The Twenty-Five Most Reusable Templates

1. Write a demand correspondence and reduce a quasilinear problem to one dimension.
2. Add two optimality inequalities to prove monotone demand.
3. Use KKT to handle state-contingent consumption with nonnegative constraints.
4. Convert endowment choice under indirect utility into value-of-endowment maximization.
5. Prove increasing differences by difference-of-differences.
6. Prove value-function convexity as supremum of affine functions.
7. Derive individual and aggregate excess demand.
8. Test continuity, homogeneity, Walras' Law, boundedness, and boundary conditions.
9. Use Walras' Law to clear the missing market.
10. Prove closed graph of an equilibrium correspondence by sequence plus continuity.
11. Draw Edgeworth box and write MRS/tangency conditions.
12. Use FWT contradiction: preferred bundles cost more, sum budgets, contradict feasibility.
13. Define strategy sets as complete contingent plans.
14. Find mixed NE by support enumeration and indifference.
15. Verify weak dominance by all-opponent-strategy inequalities.
16. Solve SPNE by subgame/backward induction and continuation values.
17. Verify sequential equilibrium with beliefs and sequential rationality.

18. Define BNE with type-contingent strategies and interim expectations.
19. Construct separating signaling equilibrium from IC inequalities.
20. Use zero-profit wages/contracts from posterior beliefs.
21. Write screening IC/IR constraints and identify monotone menus.
22. Use single crossing to prove higher type gets higher allocation.
23. Set up moral hazard with PC and IC; replace IC by FOA only if valid.
24. Use likelihood ratios to get monotone compensation and signal informativeness.
25. Use mechanism-design envelope: $U' = Q$, Q monotone, transfers from $T = xQ - U$.

19 Final Page: What to Memorize Cold

Memorize these because they are the first lines of many high-pass answers.

1. Demand correspondence: $d(p, w) = \arg \max \{u(x) : p \cdot x \leq w, x \in X\}$.
2. WE: individual optimality plus market clearing at prices.
3. PO: no feasible allocation weakly improves all and strictly improves one.
4. Walras' Law: $p \cdot z(p) = 0$.
5. Missing market: if $L - 1$ markets clear and $p_L > 0$, market L clears.
6. Strategic-form NE: each player's strategy maximizes payoff given others.
7. Strategy in extensive form: complete contingent plan at every information set.
8. SPNE: NE in every subgame.
9. SE: sequential rationality plus consistent beliefs.
10. BNE: type-by-type interim best response.
11. WPBE: strategies sequentially rational, beliefs Bayesian where possible.
12. IC menu: each type prefers its assigned contract to every other contract.
13. IR: each type weakly prefers assigned contract to outside option.
14. Single crossing proof: write two IC constraints, add, infer monotonicity.
15. Moral hazard: effort unobserved, so induce it with IC.
16. FOA caution: FOC for agent is not enough unless validity conditions hold.
17. DRM: allocation probabilities plus transfers as functions of reported types.
18. IC envelope: $U'(x) = Q(x)$ and Q is nondecreasing.
19. Revenue max: allocate to highest nonnegative virtual value.
20. Exam habit: define, set up, solve, verify, conclude.